

CS223 Computer Architecture & Organization

Introduction to DRAM System



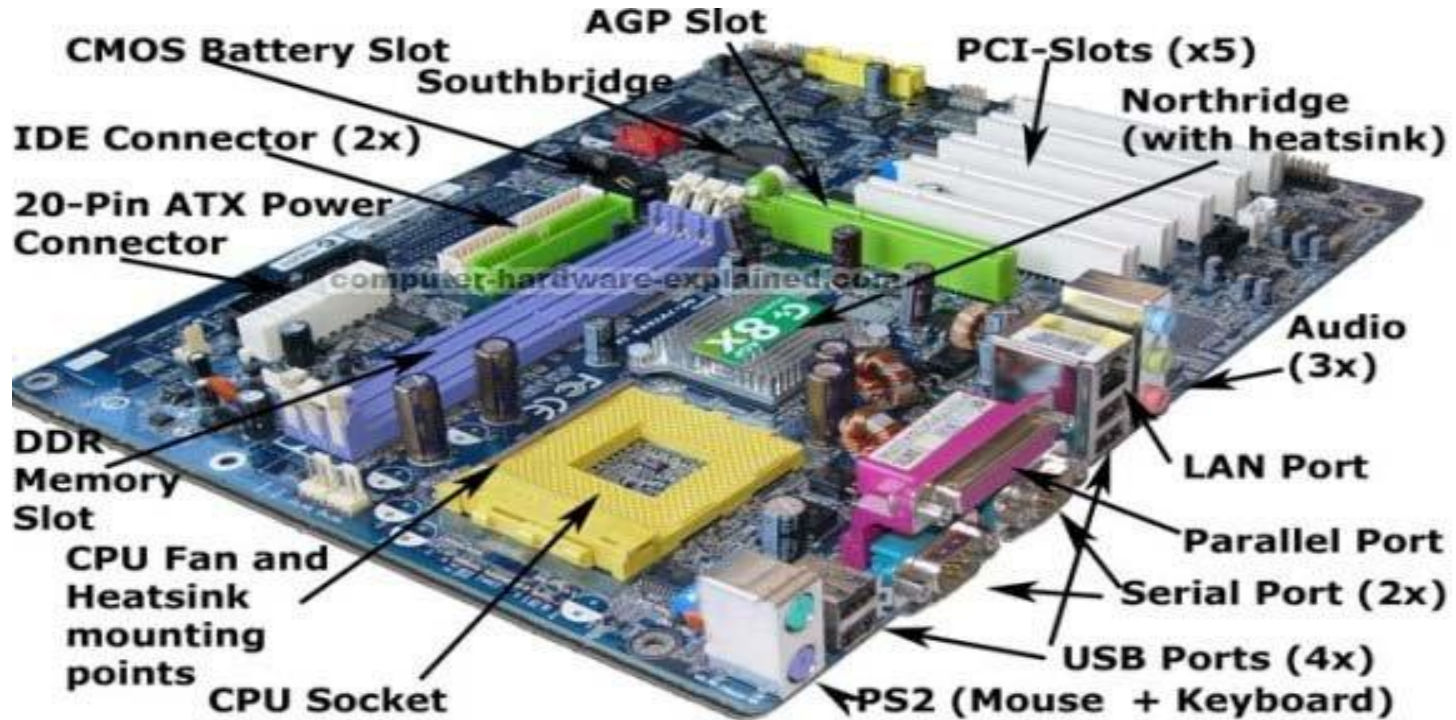
John Jose

Associate Professor

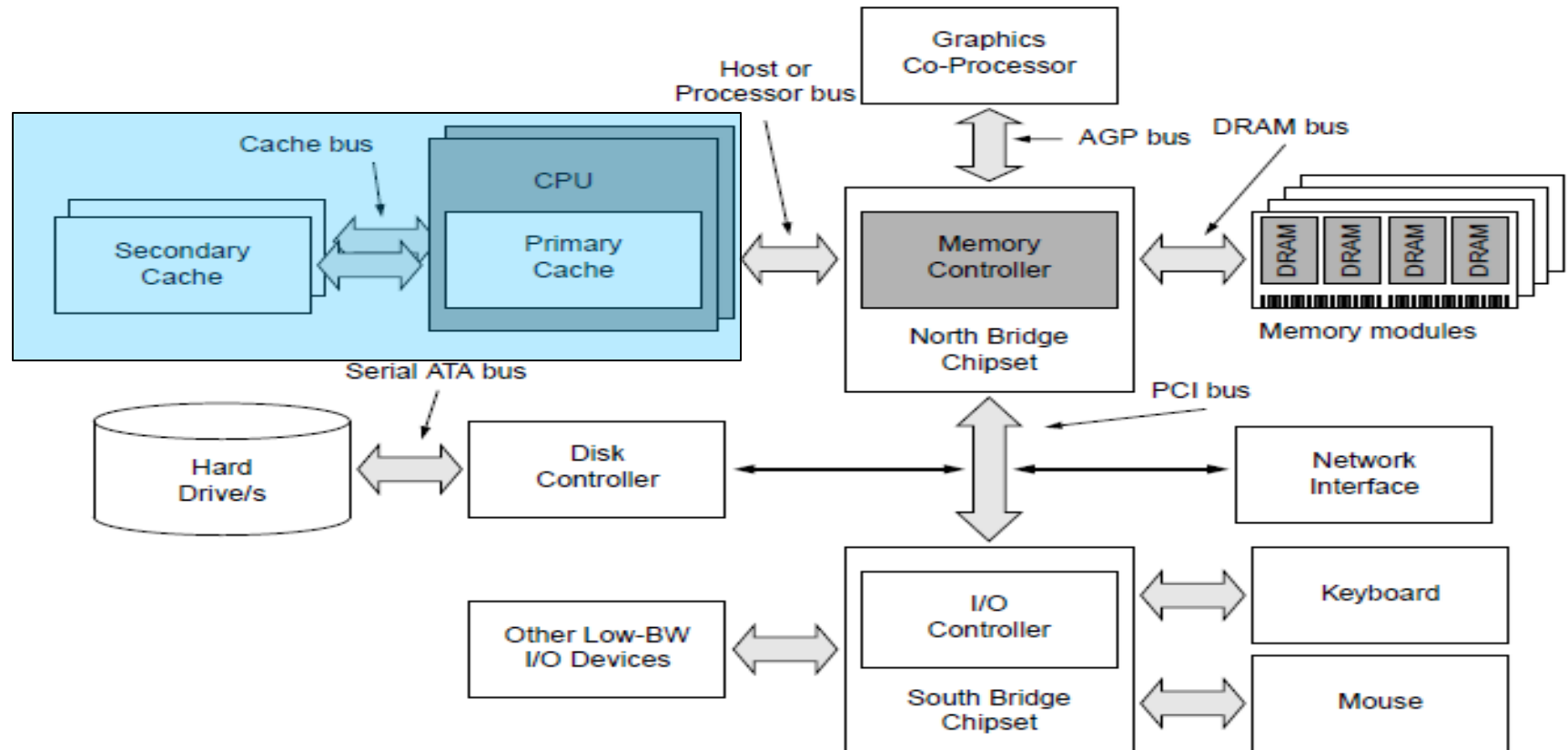
Department of Computer Science & Engineering

Indian Institute of Technology Guwahati

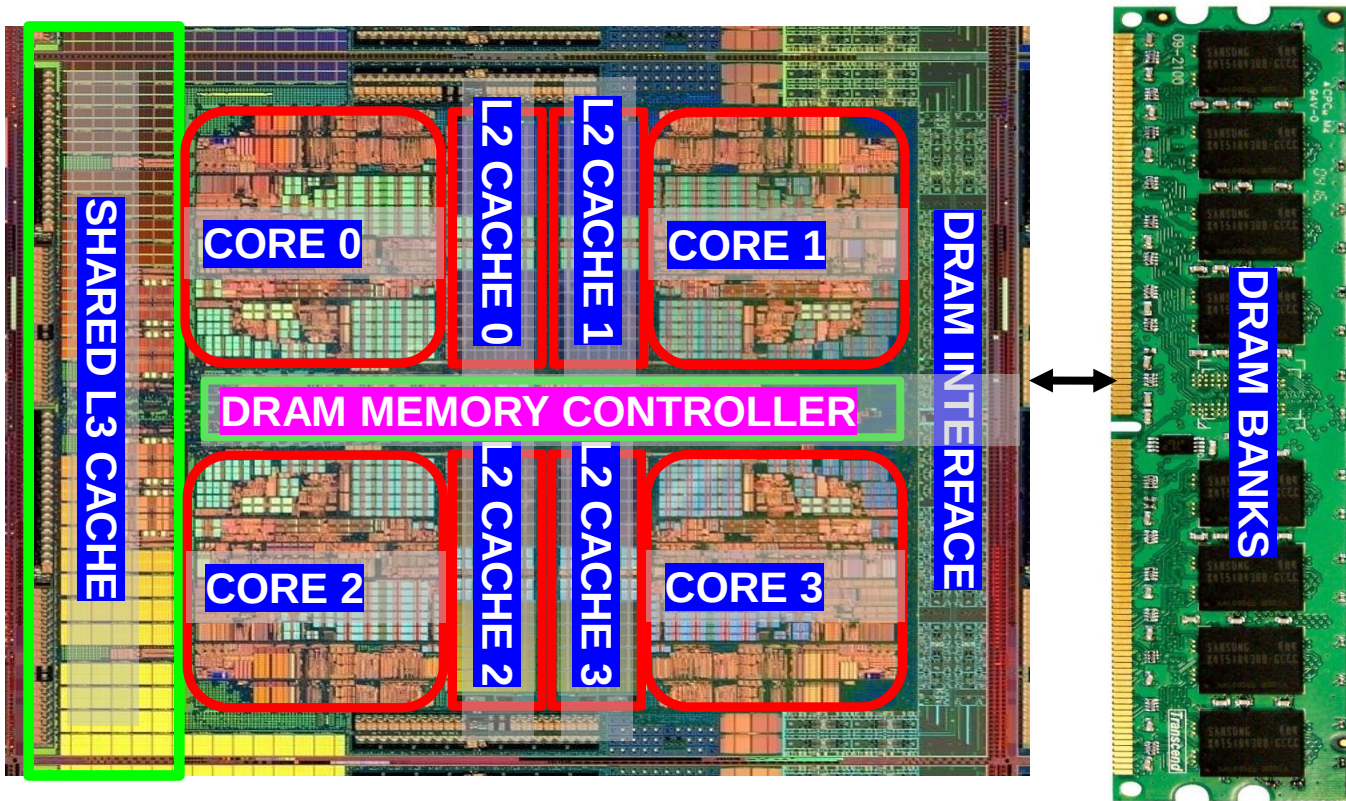
Components of a Modern Computer



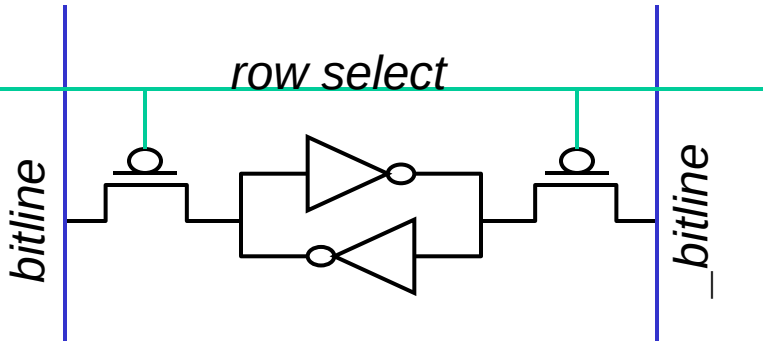
Components of a Modern Computer



Main Memory in the System



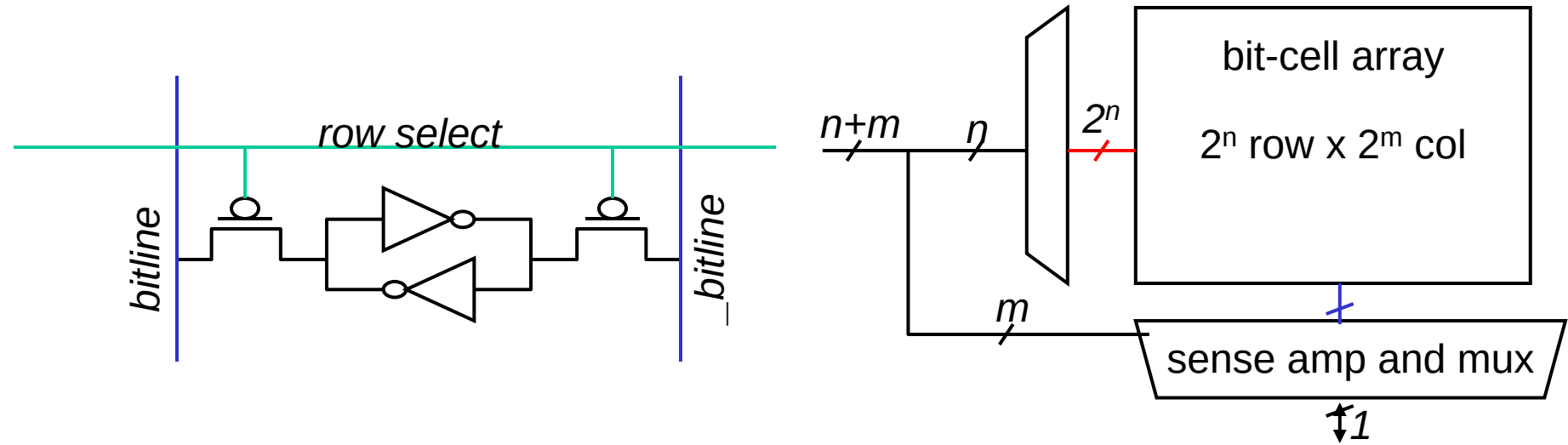
SRAM (Static Random Access Memory)



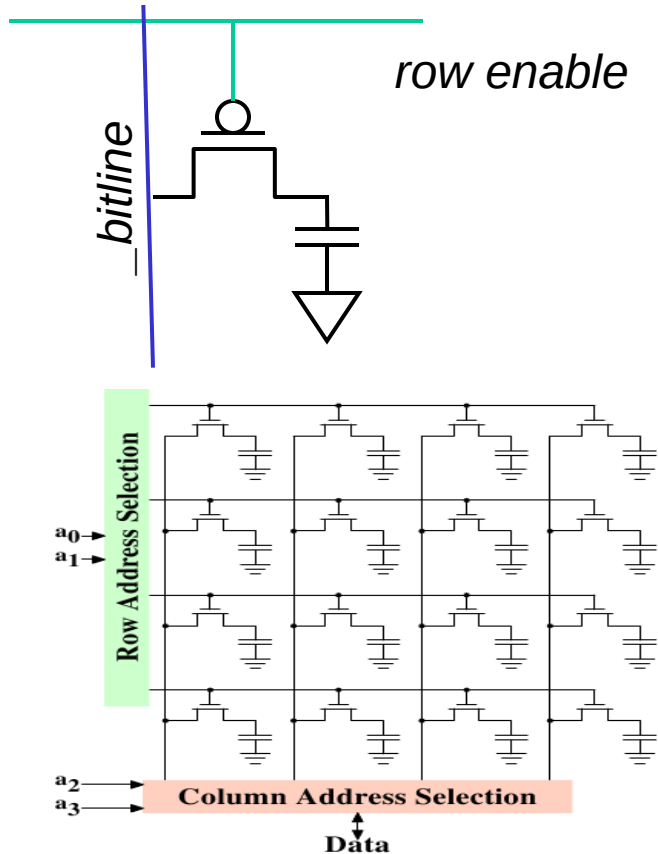
❖ Read Sequence

- ❖ address decode
- ❖ drive row select
- ❖ selected bit-cells drive bitlines (entire row is read together)
- ❖ Col. Select (data is ready)

SRAM (Static Random Access Memory)



DRAM (Dynamic Random Access Memory)



- ❖ Bits are stored as charges on capacitor
- ❖ B cell loses charge when read
- ❖ B cell loses charge over time
- ❖ A “flip-flopping” sense amplifier amplifies and regenerates the bitline, data bit is mux’ed out

DRAM vs SRAM

❖ DRAM

- ❖ Slower access (capacitor)
- ❖ Higher density (1T, 1C cell), Lower cost
- ❖ Requires refresh (power, performance, circuitry)
- ❖ Manufacturing requires putting capacitor and logic together

❖ SRAM

- ❖ Faster access (no capacitor)
- ❖ Lower density (6T cell), Higher cost
- ❖ No need for refresh
- ❖ Manufacturing compatible with logic process (no capacitor)



johnjose@iitg.ac.in
<http://www.iitg.ac.in/johnjose/>